⬡ RAVEN OS

Sovereign home security + infrastructure network · All data local · No cloud · Consent-based design

Legiom 192.168.1.6 | Duat 192.168.1.5 | Raven 192.168.1.3 | Scarab 192.168.1.2 | VPS YOUR_VPS_IP | Griffin 192.168.1.8 | *Anubis — future*

OVERVIEW

Raven OS is a self-hosted home security and infrastructure network built entirely on Raspberry Pi hardware and a Windows workstation. Every piece of data stays on your hardware. No third-party cloud services. No API keys. No signups. The network monitors files, detects threats, locks malicious content, and requires physical human confirmation to unlock — all without touching the internet for decisions.

> The network also hosts a full Egyptian-themed dungeon crawler (Gates of Darkness), a local AI (BitNet 3B), a biometric ring integration, and a WireGuard VPN with Hetzner relay — all running on the same sovereign infrastructure.

WHAT YOU BUILT

- 🔒 File threat detection + auto-lock (SHA256 → MalwareBazaar)
- 📟 Physical unlock confirmation via e-ink tap
- 🌐 LAN scanner, BLE scanner, WiFi scanner on e-ink
- 💍 Biometric ring (HR/SpO2/steps) with rolling baseline
- 🔑 WireGuard VPN — Hetzner relay, bypasses CGNAT
- 👁 Watchdog GUI — MONITOR · SLASHER · REGISTRY · OPS · SETTINGS
- ⚔ Gates of Darkness — 8-class Egyptian dungeon crawler (Medjay · Kushite · Shaduf · Sem Priest · Nubian Archer · Shardana · Heka · Tjaty)
- ⚔ 6 rarity tiers including 64 Legendary + 64 Raid items
- ⚔ Raid system — 3 bosses, multi-phase, up to 10 players
- ⚔ In-session trade system, pet AI (Scarab headless agent)
- 🤖 BitNet 3B local AI chat — no cloud, runs on CPU

NODES — HARDWARE & ROLES

LEGIOM

Main Workstation · 192.168.1.6

CPU	Intel Core i9
GPU	RTX 5070 Laptop
RAM	16 GB
OS	Windows 11 Home
SSH	OpenSSH sshd — Private network only
Auth keys	ProgramData/ssh/administrators_aut
Network	YourNetworkName — must be set to Private
WireGuard	172.16.0.6 (legiom_wireguard.conf)

Watchdog GUI · BitNet 3B LLM · Downloads monitor · Lock target (icacls)

DUAT

Home Base · Raspberry Pi 5 · 192.168.1.5 (static)

Hash DB	port 6174 · SQLite + MalwareBazaar
Unlock Service	port 6176 · SSH broker
Ring Service	port 7744 · BLE COLMI R02
Game Server	port 5000 · Gates of Darkness
Pi-hole	DNS (future)
SSH key	/home/duat/.ssh/duat_horus_key → Legiom

Always-on hub · Hash verdicts · File lock broker · Ring biometrics · Game server

RAVEN

Portable Device · Pi Zero 2 W · 192.168.1.3

Display	Waveshare 2.13" V4 · 250×122 · 1-bit
Touch	gt1151 · left/right tap
Unlock API	Flask port 6175
Heartbeat	UDP listener port 7743
SSH key	/home/raven/.ssh/pc_access_key → Legiom

E-ink OS UI · 10-item menu · Physical unlock · LAN/BLE/WiFi scanner

SCARAB

Travel Router · Pi Zero 2 W · 192.168.1.2 / 172.16.0.2

Display	Waveshare 2.13" e-ink · 250×122 · 1-bit
VPN	WireGuard · tunnel 172.16.0.2
Agent loop	scarab_agent.py · tick every 5s
Heartbeat	POST /api/pets/heartbeat every 5 min
Reports to	Duat :5000

Fiancée's router · WireGuard VPN · Headless dungeon AI · Floor retreat logic · E-ink farming display · Game pet +20% gold

VPS — HETZNER HELSINKI

ANUBIS [NOT BUILT YET]

WireGuard Relay · YOUR_VPS_IP

Purpose	Bypasses T-Mobile CGNAT
WireGuard port	UDP 443 (disguised as HTTPS)
Server pubkey	YOUR_SERVER_PUBKEY
DDNS	YOUR_DDNS_DOMAIN → YOUR_VPS_IP

`Always-on relay` `External VPN gateway` `No data stored`

Son's Guardian Node · Pi Zero 2 W

Purpose	Vitals + location monitoring
Consent	"They must hold hands" — both rings physically present
Handshake gate	Stress detected → connection blocked
Revoke	Either party · unilateral · instant
Game hook	Low HR → harder dungeon enemies

`Future` `Consent-gated`

COLMI R02

Biometric Ring · the user's right hand

BLE address	XX:XX:XX:XX:XX:XX
Device name	YOUR_RING_NAME
Data	HR · SpO2 · steps · battery
Reads via	Duat Ring Service port 7744
Baseline	Rolling confidence score — readiness, not a gate
Gestures	Iron House edit context only
2nd ring	Hook in raven_ring.py · fiancée · future

WIREGUARD VPN

> All external traffic routes through Hetzner VPS at `YOUR_VPS_IP:443` . T-Mobile CGNAT blocks inbound connections to Duat — the VPS relay is the bypass. All clients use full-tunnel routing (`AllowedIPs = 0.0.0.0/0`) and Pi-hole DNS (`192.168.1.5`).

CLIENT CONFIGS (verified — all on Desktop)

File	Device	Tunnel IP	Endpoint

pixel_wireguard.conf	user's phone	172.16.0.5/32	YOUR_VPS_IP:443
legiom_wireguard.conf	Legiom (this machine)	172.16.0.6/32	YOUR_VPS_IP:443
mandy_wireguard.conf	fiancée's iPhone	172.16.0.3/32	YOUR_VPS_IP:443
RavenOS/Scarab/aaron_phone.conf	Pixel (alternate)	172.16.0.5/32	YOUR_VPS_IP:443

Server public key: `YOUR_SERVER_PUBKEY`

All configs: `PersistentKeepalive = 25`, `DNS = 192.168.1.5`

PHONE HOTSPOT TUNNEL SHARING

When user's phone runs WireGuard and creates a hotspot, devices on that hotspot share the tunnel automatically — connecting to Duat via VPN without needing their own WireGuard install. The phone becomes a relay for tethered devices.

TUNNEL IP MAP

IP	Device
172.16.0.2	Scarab (travel router)
172.16.0.3	fiancée's iPhone
172.16.0.5	user's phone
172.16.0.6	Legiom

STATUS & PENDING

- ✓ Hetzner VPS running at YOUR_VPS_IP
- ✓ All 4 client configs generated and on Desktop
- ✗ Netgear port forward — change "WireGuard" rule from TCP → UDP (port 51820) in router admin
- ✗ DDNS crontab on Duat — `YOUR_DDNS_DOMAIN` must update to VPS IP
- ✗ Legiom Private network — run `Set-NetConnectionProfile -Name "YourNetworkName" -NetworkCategory Private`

DIAGNOSTICS

```
sudo wg show
sudo wg show wg0 latest-handshakes    # handshake within 3 min = alive
```

```
sudo wg-quick down wg0 && sudo wg-quick up wg0   # restart
nslookup YOUR_DDNS_DOMAIN            # must resolve to YOUR_VPS_IP
```

SERVICES & PORTS

Port	Proto	Service	Host	Commands
5000	TCP	Gates of Darkness game server	Duat	`sudo systemctl restart raven-slasher`
6174	TCP	Hash DB API	Duat	`sudo systemctl restart duat-hashdb`
6175	TCP	Raven Unlock API	Raven	Started by raven_deck.py
6176	TCP	Duat Unlock Service	Duat	`sudo systemctl restart duat-unlock`
6177	TCP	Watchdog Unlock Listener	Legiom	Built into watchdog_windows.py
7743	UDP	Heartbeat listener	Raven	Built into raven_deck.py
7744	TCP	Ring biometric API	Duat	`sudo systemctl restart raven-ring`
7744	UDP	Scarab heartbeat listener	Legiom	Built into watchdog_windows.py
8080	TCP	BitNet LLM chat	Legiom	start_chat.bat (localhost only)
22	TCP	SSH	All nodes	—
51820	UDP	WireGuard	VPS + Duat	Router: port forward must be UDP (not TCP)
443	UDP	WireGuard (disguised)	VPS	VPS endpoint port — bypasses firewalls

HEALTH CHECKS

```
curl http://192.168.1.5:6174/health     # Hash DB
curl http://192.168.1.5:6174/stats
curl http://192.168.1.5:6176/health     # Unlock service
curl http://192.168.1.5:7744/status     # Ring service
curl http://192.168.1.5:7744/biometrics/latest
curl http://192.168.1.5:5000            # Gates of Darkness
curl http://192.168.1.5:5000/api/ladder # Leaderboard JSON
```

DEPLOY GAME SERVER

```
scp C:/Users/YourUser/Desktop/app.py duat@192.168.1.5:/opt/raven-slasher/app.py
scp C:/Users/YourUser/Desktop/ladder_slasher.html duat@192.168.1.5:/opt/raven-slasher/ladder_slasher.htm
ssh duat@192.168.1.5 "sudo systemctl restart raven-slasher"
```

PI-HOLE — NETWORK DNS SINKHOLE

Pi-hole runs on Duat (Pi 5), serving as the DNS resolver for all LAN clients. All WireGuard clients also use Pi-hole DNS (`192.168.1.5`) — they get ad blocking regardless of where they are. Future deployment.

> All LAN clients and all WireGuard clients point to `192.168.1.5` as DNS. WireGuard configs already have this baked in.

SERVICE DETAILS

Host	Duat — Pi 5 · 192.168.1.5
DNS port	53 TCP/UDP
Web admin	http://192.168.1.5/admin
Status	Future — not yet deployed

CONFIGURATION

Upstream DNS	1.1.1.1 or 9.9.9.9 (no Google)
Local DNS	duat.local · raven.local · legiom.local
WG clients	All point to 192.168.1.5 already

MANAGEMENT

```
pihole status              # check running + blocking
pihole -t                  # tail query log live
pihole restartdns           # restart resolver
pihole -g                  # update gravity (blocklists)
pihole disable 5m          # temp disable
```

> Always set secondary DNS to 1.1.1.1 in router DHCP. If Pi-hole goes down and no secondary is set, all LAN clients lose internet.

SSH ARCHITECTURE

From	To	Key	Purpose
Raven	YourUser@Legiom	/home/raven/.ssh/pc_access_key	Remote Downloads scan
Duat	YourUser@Legiom	/home/duat/.ssh/duat_horus_key	Lock/unlock files (icacls)
YourUser	raven@Raven	C:\Users\YourUser\.ssh\	Deploy + manage Raven
YourUser	duat@Duat	C:\Users\YourUser\.ssh\	Deploy + manage Duat

> YourUser is a Windows Administrator account. Windows OpenSSH ignores per-user authorized_keys for admin accounts. Use the administrators_authorized_keys path only.

LEGIOM AUTHORIZED KEYS

Location: `C:\ProgramData\ssh\administrators_authorized_keys`

Permissions: `NT AUTHORITY\SYSTEM:F` and `BUILTIN\Administrators:F` only — OpenSSH enforces this strictly. Any extra ACL entry breaks auth.

Contains: raven-os-remote-access · duat@duat · duat@duat-rotated · scarab@scarab-rotated · raven@raven-rotated

SSHD_CONFIG

```
# C:\ProgramData\ssh\sshd_config — must contain (uncommented):
AuthorizedKeysFile __PROGRAMDATA__/ssh/administrators_authorized_keys
```

FIREWALL

```
# Run once in elevated PowerShell:
Set-NetConnectionProfile -Name "YourNetworkName" -NetworkCategory Private
```

FILE LOCK / UNLOCK FLOW

```
1. File lands in Downloads (Legiom)
↓
2. Watchdog detects via folder watcher → SHA256 hash
↓
3. GET https://192.168.1.5:6174/lookup/:sha256
```

```
↓ SQLite hit → verdict · Cache miss → CIRCL hashlookup · Unknown → submit to MalwareBazaar
anonymously
4. Verdict returned
├── CLEAN → file allowed, no action
└── MALICIOUS ↓
5. POST lock request → Duat :6176
↓
6. Duat SSHes as YourUser@Legiom → icacls /deny YourUser:RX
↓
7. Duat forwards alert → POST Raven :6175
↓
8. Raven e-ink: THREAT DETECTED · tap to dismiss → UNLOCK QUEUE
↓ Navigate: Menu → Unlock Queue → Select file → UNLOCK CONFIRM
9. Tap UNLOCK (left) or DENY (right)
↓
10. Raven POSTs decision → Duat :6176
↓
11. UNLOCK: Duat SSHes → icacls /grant YourUser:F → file accessible
DENY: File remains locked on disk
```

> Watchdog also listens on port 6177 for direct unlock instructions. Local icacls lock/unlock via `icacls /deny` and `icacls /remove:d + /grant`.

WATCHDOG GUI — LEGIOM

File: `watchdog_windows.py` — Tkinter GUI. Config: `%APPDATA%\RavenOS\watchdog_config.json`

BACKGROUND THREADS (always running)

Thread	Behavior
Heartbeat loop	UDP `WATCHDOG|Legiom|YourUser|ALIVE` → Raven :7743 every 30s
Scarab listener	UDP port 7744 — receives Scarab heartbeat packets
Unlock listener	HTTP port 6177 — receives /unlock and /unlock_decision
Process monitor	60-second scan via `process_monitor` module (requires psutil)
Status loop	Polls Duat/Raven health every 30s

Slasher loop	Polls /api/ladder every 15s for SLASHER tab

CONFIG VALUES

Setting	Default
downloads_path	C:\Users\YourUser\Downloads
duat_ip	192.168.1.5
duat_port (Hash DB)	6174
duat_unlock_port	6176
raven_ip	192.168.1.3
raven_port	6175
raven_heartbeat_port	7743

TAB: MONITOR

Core security panel. Folder watcher (watchdog library) detects new files in Downloads. SHA256 hash → queries Duat Hash DB over HTTPS. Shows: files scanned / threats / unknown / locked. Scrollable event log with verdicts. Suspicious extensions flagged: `.exe .bat .cmd .ps1 .vbs .js .msi .dll .scr .pif .com .jar .hta .wsf .reg .inf`.

TAB: SLASHER

Header: **"GATES OF DARKNESS"**. Live leaderboard from Duat port 5000. 🎮 PLAY button opens `http://192.168.1.5:5000`. Columns: rank · player · level · floor · kills · class · name. Auto-refresh every 15s. Rank badges: ★ top 3 · ◆ top 10 · ◉ rest.

TAB: REGISTRY

Network Key Registry. 5 node selector buttons (RAVEN · DUAT · LEGIOM · SCARAB · ANUBIS). Per-node: connection config panel (IP/port/role) and public key panel (add/remove SSH keys). Trusted/Pending counts in header. Persisted to `%APPDATA%\RavenOS\registry.json`.

TAB: OPS

OPS Console. QUICK SCAN button fires all 8 checks in parallel. Each row: subsystem name · status dot (●) · STATUS label · LAST CHECK timestamp · individual CHECK button. Subsystems monitored:

Subsystem	Check Method
WireGuard Tunnel	SSH → `sudo wg show interfaces`
DDNS	SSH → `crontab -l | grep no-ip`

Heartbeat System	UDP ping to Duat :7744
SSH Key Auth	BatchMode SSH to each node
Pi-hole DNS	SSH → `pihole status`
Watchdog GUI	Self-check (always passes)
Duat Game Server	HTTP GET /api/ladder
Remote Control Chain	Manual user-verified

TAB: SETTINGS

Downloads Folder path · Duat IP / Hash DB Port / Unlock Port · Raven IP / Port · Checkboxes: Enable hash DB lookups / Auto-lock threats / Start watching on launch. TEST CONNECTIONS button. SAVE button.

RAVEN OS — E-INK MENU

> Touch controls: Left tap = cycle/action · Right tap = select/back · Main screen left tap = inject test unlock alert

```
MAIN
└── MENU
    ├── 1.  LAN PING SWEEP      — nmap ping scan 192.168.1.0/24
    ├── 2.  VIEW LAN INTEL      — discovered nodes + open ports
    ├── 3.  AUTO-AUDIT LAN      — port scan all live hosts
    ├── 4.  BLUETOOTH           — BLE scan, nearby devices
    ├── 5.  WIFI APs            — scan SSIDs + signal strength
    ├── 6.  SYS HEALTH          — CPU temp, mem, disk, uptime
    ├── 7.  VIEW LOGS           — tail raven logs, scrolling
    ├── 8.  REMOTE SCAN PC      — SSH to Legiom, scan Downloads
    ├── 9.  UNLOCK QUEUE        — pending lock alerts, tap to resolve
    ├── 10. SHUT DOWN           — sudo shutdown -h now
    └── <- BACK TO MAIN
```

HEARTBEAT PROTOCOL

Watchdog sends `WATCHDOG|Legiom|YourUser|ALIVE` UDP to Raven :7743 every 30 seconds. Raven listens and updates display status. Timeout > 90s → LEGIOM OFFLINE alert on e-ink.

SCARAB — HEADLESS DUNGEON AGENT

Scarab runs `scarab_agent.py` as a headless pet AI. State persisted at `/tmp/scarab_pet_state.json`. Displays combat status on Waveshare 2.13" e-ink.

AGENT LOOP

```
Main loop — tick every 5s
├── Post heartbeat every 5 min (→ Duat :5000 /api/pets/heartbeat)
└── Fight loop:
Roll hit · pet ATK vs mon DEF → damage
├── Monster dies → XP + gold + loot · sync to Duat
└── Pet dies → consecutive_deaths++
├── deaths ≥ 2 AND floor > 1 → RETREAT: floor-1, deaths reset
└── otherwise → respawn floor 1 at 60% HP
```

WIREGUARD — MANDY'S VPN

Scarab acts as fiancée's travel router. Config generated (`mandy_wireguard.conf`), QR code ready. **Blocked:** Netgear port forward must be TCP → UDP. iPhone "Maximize Compatibility" must be ON to force 2.4 GHz (Pi Zero 2 W is 2.4 GHz only).

GAME PET BONUS

When Scarab is online (heartbeat active on Duat): +20% gold on all kills. Combined with Raven online: +8% XP bonus.

OPS — WATCHDOG SUBSYSTEM HEALTH

The OPS tab provides a live health dashboard. QUICK SCAN fires all 8 checks in parallel threads, updating status dots without blocking the UI.

TRIAGE WORKFLOW

```
1. Run Quick Scan — all 8 section checks fire in parallel
↓
2. Failure found on a section
↓ Go to that section's individual CHECK button
```

```
3. Individual check fails → escalate to full diagnostic
4. All checks pass but system still broken?
↓ Check Ledger Sync — verify Duat has the current MD
PUSH TO ALL → re-verify → recheck
```

LAST KNOWN GOOD STATE

Stored at: `%APPDATA%\RavenOS\ops_last_good.json` — snapshot of last passing state. MARK GOOD button saves current state as baseline.

RING SERVICE — COLMI R02

Daemon: `raven_ring.py` on Duat. BLE address: `XX:XX:XX:XX:XX:XX` (device name `YOUR_RING_NAME`). worn on the right hand.

API ENDPOINTS

Endpoint	Returns
/status	Ring connected, battery, last seen
/biometrics/latest	HR, SpO2, steps
/baseline	Rolling baseline + confidence score
/gesture	Last detected gesture (Iron House context only)

DEPLOY

```
scp raven_ring.py raven_ring.service duat@192.168.1.5:/home/raven/
ssh duat@192.168.1.5 "sudo cp /home/raven/raven_ring.service /etc/systemd/system/ && sudo systemctl enab
pip install colmi-r02-client --break-system-packages
python3 raven_ring.py --test-gestures  # gesture calibration
```

> Confidence score = personal readiness indicator, not a system gate. Gesture detection only fires in Iron House editing context. Hook left for second ring (fiancée, left hand) — not built yet.

GATES OF DARKNESS

Egyptian-themed dungeon crawler. Server: `app.py` (Flask + SQLite, WAL mode) on Duat port 5000. Client: `ladder_slasher.html` served by Flask. Play at `http://192.168.1.5:5000`. DB: `/opt/raven-slasher/ladder_slasher.db`.

CLASSES (8 — all verified in code)

Key	Lore Name	Primary	Passive
fighter	Medjay 🗡️	STR · VIT	Counter-Strike: 25% chance to deal 40% dmg back when hit
barbarian	Kushite 🪓	STR	Bloodlust: +8% dmg per hit taken (max 5 stacks), unleash on next attack
rogue	Shaduf 🥷	DEX	Ambush: first strike vs each new enemy = guaranteed crit
mage	Sem Priest 🧙	INT	Arcane Body: all damage scales INT, skills ignore 50% armor
ranger	Nubian Archer 🏹	DEX	Keen Eye: always acts first, first hit each fight = guaranteed crit
samurai	Shardana ⚔️	STR · DEX	Death Blow: 2× damage when enemy HP < 30%
cleric	Heka ✨	INT · VIT	Divine Light: healing skills +20% HP restored, holy damage ignores 30% armor
paladin	Tjaty ⚜️	STR · VIT	Bulwark: 15% less damage taken; heals grant a shield (10% of heal amount)

SKILLS PER CLASS (all verified in ladder_slasher.html)

Class	L1	L5	L10	L15
Medjay	Iron Strike (150% STR)	War Cry (−25% enemy ATK)	Shield Bash (stun + 80%)	Last Stand (200%, heal if <30%)
Kushite	Rampage (180% STR)	Berserker's Howl (+40% ATK)	Cleave (2× 90%)	Primal Rage (300%, costs 10% HP)
Shaduf	Serpent Strike (220% DEX)	Hemorrhage (5% HP/turn)	Shadow Step (dodge + counter)	Death Mark (3× if <40%)
Sem Priest	Sekhmet's Wrath (INT×3)	Frost Chains (−40% ATK)	Soul Drain (INT×2 + MP)	Wrath of Ra (INT×6 AOE)

Nubian Archer	Arrow Storm (DEX×1.6)	Piercing Shot (DEX×2, armor)	Rain of Arrows (3× 70%)	Eagle Eye (guaranteed crit 250%)
Shardana	Dual Onslaught (2.5× total)	Blade Flash (120%, CD1)	Iaijutsu (300%)	Void Cut (25% current HP)
Heka	Searing Light (INT×2.5 holy)	Mend Wounds (heal 30% HP)	Purifying Flame (AOE + heal 10%)	Breath of Osiris (heal 60% + 3-hit shield)
Tjaty	Smite (160% STR holy)	Lay on Hands (heal 20% + shield)	Holy Shield (−40% dmg, 3 hits)	Judgment (massive holy, heal 50% dmg)

EQUIPMENT SLOTS (8)

`weapon` · `armor` · `helmet` · `ring` · `belt` · `boots` · `gloves` · `amulet`

RARITY TIERS (6 — all verified in code)

Tier	Color	Notes
Common	Gray	Basic stats, single slot
Uncommon	Blue	Better stats, some secondaries
Rare	Purple	Multi-stat gear
Unique	Gold	Named items, floor-scaled random stats via `generate_unique(floor)`, special effects
Legendary	Orange	64 items total (8 per class), fixed stats, level req 15, untradeable, class affinity, passive enhancement effects
Raid	Red	64 items (8 per class), drop from raid bosses only, `tradeable:False`, stronger than Legendary, on-hit procs

STATUS EFFECTS (5 — all verified in monster pool)

Effect	Applied By
poison	Sand Viper, Cobra Guardian, Scorpion Swarm, Plague Locust, Son of Apophis, various raid items
bleed	Tomb Spider, Abyssal Scorpion, various raid weapons
burn	Deshret Demon, Chaos Elemental, Ra's Fury, Chaos God Fragment, various raid items
frostbite	Thoth's Riddle, Staff of the First Word (raid weapon)
curse	Cursed Scribe, Soul Eater, Nephthys Shade, Eternal Wraith, The Unnamed God, various raid items

LIFE STEAL / MANA STEAL (verified in Unique items)

Life steal: **Blade of the Dusk** — "heal 8% of damage dealt on each hit" · Mana steal: **Staff of Set's Gale** — "Restore 5 MP on every kill"

TRADE SYSTEM (verified – fully implemented)

In-session co-op trade between two players. Server-side with offer/accept/cancel flow.

```
GET  /api/sessions/<sid>/trade         – view current offers
POST /api/sessions/<sid>/trade/offer   – place offer
POST /api/sessions/<sid>/trade/accept  – accept offer
POST /api/sessions/<sid>/trade/cancel  – cancel offer
```

ROOM TYPES (verified in dungeon generator)

Type	Description
enemy	Monster + loot on kill
loot	Guaranteed item drop
trap	DEX-based disarm. Disarm=loot, spring=damage+no loot, skip=safe
shrine	Spend gold for HP or MP restoration (scales with floor)
merchant	Sells Rejuvenation Potions (35% HP + 35% MP)
empty	May have rest point (12.5% roll on first entry)
boss	Furthest from start, drops Rare+ loot, unlocks stairs on death
stairs	Locked until boss dies, descend to next floor
roulette	Gamble room — verified in generator + `/api/sessions/<sid>/roulette`
raid_boss	Special boss room on raid floors — triggers full raid encounter

PETS SYSTEM (verified – full implementation)

DB tables: `pets`, `pet_bindings`. Pets register via device IP (Scarab), have class/icon/stats/level/XP. Characters claim pets with PIN. Pets deliver loot, send heartbeats, participate in dungeon play sessions.

```
GET  /api/pets/bound           – pets bound to character
GET  /api/pets/status          – all pet statuses
POST /api/pets/claim           – claim pet by PIN
POST /api/pets/unbind          – unbind from character
GET  /api/pets/bindings        – all bindings
POST /api/pets/deliver         – deliver pending loot
POST /api/pets/heartbeat       – heartbeat + state sync
```

```
GET  /api/pets/active_heroes   — heroes active in last N min
POST /api/pets/play/sync       — pet play session sync
```

RAID SYSTEM (verified — 3 bosses, multi-phase)

Boss	Difficulty	HP/Player	Phase Mechanics
Warden of the Gates ⚖️	3	700	Gate Stance → Judgment (mark) → Shadow Split (adds) → Frenzy
Apophis the Devourer 🐍	4	1000	Coiling → Venom Surge (poison pool) → Devour → Chaos Unbound (shield)
The Pharaoh Ascendant 👑	5	1400	Divine Reign → Plague of Gods → Wrath and Ruin (adds DPS check) → Rebirth

```
POST /api/raids                  — create lobby
GET  /api/raids/open             — list open raids
POST /api/raids/<rid>/join       — join raid
GET  /api/raids/<rid>            — poll state
POST /api/raids/<rid>/ready      — mark ready
POST /api/raids/<rid>/start      — start (host only)
POST /api/raids/<rid>/attack     — attack action
POST /api/raids/<rid>/attack_add — attack adds/mechanic
POST /api/raids/<rid>/leave      — leave raid
```

MONSTER ROSTER (by floor tier — verified)

Tier	Floors	Zone	Notable Examples
1	1–5	Tomb Entrance	Tomb Scarab, Desert Rat, Sand Viper (poison), Grave Robber, Clay Golem
2	6–10	Catacombs	Mummified Priest, Cursed Scribe (curse), Bone Stalker, Tomb Spider (bleed)
3	11–15	Underworld Gates	Ammit Spawn, Deshret Demon (burn), Soul Eater (curse), Obsidian Golem
4	16–20	Duat Depths	Apep Serpent (poison), Set Beast, War Sphinx, Chaos Elemental (burn)
5	21–25	Divine Realm	Son of Apophis, Ra's Fury (burn), Sekhmet Avatar, Anubis Judge
6	26+	Infinite Abyss	Mythic variants — Void Osiris, Duat Titan, The Unnamed God (curse)

COMBAT CONTROLS

`Space` = Attack · `1/2/3/4` = Skills · `F` = Flee · `H` = Potion · `W/A/S/D` / arrows = Move · `R` = Rest

MOBILE

Swipe zones: `.sz-left` and `.sz-right` fixed to bottom of screen, full-width (50% each) on mobile. All game screens responsive via CSS media queries. Game title element in HTML: `Gates of Darkness`.

CO-OP / SPECTATE / CHAT (verified)

Co-op: 2 players, server-authoritative dungeon, same map, boss HP scales, polls every 2.5s. Spectate any online player → 5×5 minimap + event log (2.5s poll). Global chat: 💬 toggle, all screens, polls every 5s, 200 char limit.

ALL API ENDPOINTS

```
POST /api/register · POST /api/login · GET /api/users
GET/POST /api/characters · PUT /api/characters/<cid> · POST /api/characters/<cid>/delete
GET /api/ladder · GET /api/stats · GET /api/online
GET/POST /api/chat
POST /api/sessions · GET /api/sessions/open
GET /api/sessions/<sid> · POST .../move · POST .../sync · POST .../attack
POST .../loot · POST .../buy · POST .../roulette · POST .../update_player
POST .../new_floor · POST .../close
GET/POST/POST/POST /api/sessions/<sid>/trade[/offer|/accept|/cancel]
[Pets: see Pets section above]
[Raids: see Raid section above]
GET /  → ladder_slasher.html
```

LEADERBOARD

Sorted alive-first, then by floor/kills. Rank badges: ★ top 3 · ◆ top 10 · ◉ rest. SLASHER tab auto-refreshes every 15s.

BITNET — LOCAL AI CHAT

Microsoft's 1.58-bit LLM inference running entirely on CPU on Legiom. No cloud. No API key. Future — installer ZIP ready.

Active Model

Model	Falcon3-3B-Instruct-1.58bit
Format	I2_S GGUF · 2.1 GB on disk
Speed	~20 tokens/sec · 8 threads

Files

Launch	C:\Users\YourUser\BitNet\BitNet\sta
Model file	...\models\Falcon3-3B-Instruct-1.58bit\ggml-model-i2_s.gguf
Rebuild	C:\Users\YourUser\Desktop\build_bi

Chat UI	http://localhost:8080
API	OpenAI-compatible /v1/chat/completions
Build stack	clang-cl · CMake · VS 2022 · BITNET_X86_TL2=ON

BUILD BACKLOG

RAVEN OS INFRASTRUCTURE

- ✓ Watchdog GUI — file hashing, lock/unlock, heartbeat, 5 tabs
- ✓ Duat Hash DB — MalwareBazaar + CIRCL + SQLite
- ✓ Duat Unlock Service — SSH broker
- ✓ Raven e-ink OS — 10-item menu, touch input
- ✓ Raven Flask API — unlock queue
- ✓ Scarab dungeon agent — headless pet AI, floor retreat logic
- ✓ Scarab e-ink farming display — combat aura, event log
- ✓ Ring service — COLMI R02 BLE, biometrics, gesture engine
- ✓ OPS tab — subsystem health grid, 8 parallel checks
- ✓ Registry tab — SSH key management per node
- ✓ WireGuard VPN — Hetzner relay, 4 client configs generated
- ✓ BitNet local AI — Falcon 3B instruct on Legiom
- ✓ Legiom SSH firewall — Set-NetConnectionProfile (elevated PowerShell)
- ✗ WireGuard port forward — change TCP → UDP in Netgear admin
- ✓ MalwareBazaar parser fix — fix_hashdb_parser.sh deployed
- ✓ Raven deck patch — patch_raven_deck.sh deployed
- ✗ Scarab WiFi — enable "Maximize Compatibility" on iPhone, verify connect
- ✗ DDNS setup — No-IP crontab on Duat, point YOUR_DDNS_DOMAIN to VPS IP
- ✗ Pi-hole DNS on Duat
- ✗ Duat SSH hardening — disable password auth, key-only
- ✗ Process monitor — pip install psutil on Legiom
- ○ Anubis — son's guardian node, "hold hands" consent mechanism

- ○ Fiancée's second COLMI ring (hook in raven_ring.py)

GATES OF DARKNESS GAME BACKLOG

- ✓ 8 classes — Medjay, Kushite, Shaduf, Sem Priest, Nubian Archer, Shardana, Heka, Tjaty
- ✓ 8 equipment slots (weapon/armor/helmet/ring/belt/boots/gloves/amulet)
- ✓ 6 rarity tiers (Common → Raid)
- ✓ 64 Legendary items (8 per class, class affinity, untradeable)
- ✓ 64 Raid items (8 per class, raid boss drops, untradeable)
- ✓ 3 Raid bosses with multi-phase mechanics
- ✓ 5 status effects (poison/burn/bleed/frostbite/curse)
- ✓ Life steal / mana steal (Unique items)
- ✓ In-session trade system
- ✓ Pet system (pets DB table + bindings, heartbeat, loot delivery)
- ✓ Roulette room type
- ✓ Raid boss room type
- ✓ Mobile swipe zones
- ✓ 6-tier monster roster (floors 1-5 through 26+)
- ✓ Co-op, spectate, global chat
- ○ Co-op skill sync — server-side skill effects (currently simplified)
- ○ Sound effects — not yet in client
- ○ Floor balance review — enemy stats at floors 5+
- ○ Vault room — locked chest, key from elites
- ○ Forge room — upgrade equipment
- ○ Skill tree branching — 2-option choices at each milestone
- ○ Anubis integration — low HR → harder enemies
- ○ Second ring (fiancée) → dungeon events

PHILOSOPHY & KEY CONCEPTS

SOVEREIGN INFRASTRUCTURE

All data stays local. No third-party cloud. No API keys. No signups. Community feeds (MalwareBazaar) used anonymously. Unknown hashes submitted back anonymously — contributing without identifying. You own the hardware, you own the data.

SECURITY BY DESIGN

Authentication, encryption, and input sanitization built in by default — not added on request. Physical confirmation required for file unlock. No remote software bypass. The unlock chain requires a human hand on the device.

CONSENT-BASED DESIGN

"They have to hold hands." — The Anubis consent mechanism. Both rings must be physically present simultaneously to establish a monitoring connection. Cannot be initiated remotely. Vitals checked during handshake — if stress detected, connection blocked. Either party can revoke unilaterally. This is the ethical foundation of any future product.

SELF-HEALING NETWORK

Scarab detects Duat offline → falls back gracefully → reconnects when Duat returns. Fiancée never notices. Zero friction. The network heals around failures silently.

THE RAVEN RING

The COLMI R02 (right hand). Biometrics feed a rolling baseline — confidence score is a personal readiness indicator, not a system gate. Gesture vocabulary isolated: Iron House editing gestures only fire in editing mode. Hook left for fiancée — not built yet.

NETWORK REFERENCE

Item	Value
Router	Netgear Nighthawk R6700
Subnet	192.168.1.0/24
Duat	192.168.1.5 (static)
Raven	192.168.1.3
Scarab	192.168.1.2 LAN / 172.16.0.2 WG
Legiom	192.168.1.6
VPS (Hetzner)	YOUR_VPS_IP
WireGuard tunnel	172.16.0.0/24

WG server pubkey	YOUR_SERVER_PUBKEY